Power System Voltage Stability Evaluation Considering Renewable Energy With Correlated Variabilities

Xiaoyuan Xu, *Member, IEEE*, Zheng Yan, Mohammad Shahidehpour, *Fellow, IEEE*, Han Wang, and Sijie Chen, *Member, IEEE*

***Abstract*—Power system operation will encounter numerous variabilities as the proliferation of renewable energy continues. Such random events will require the evaluation of corresponding variables and the assessment of their impacts on the monitoring and the control of stochastic power systems. In this paper, a global sensitivity analysis (GSA) method is proposed to perform a priority ranking of renewable energy variabilities that will affect the voltage stability of power systems. First, a probabilistic model for the load margin calculation is presented considering the renewable power generation. Then, GSA is applied to models with correlated random input variables and the importance index is designed for evaluating the impacts of such variables. Next, the stochastic response surface method is adopted in GSA to improve the computation efficiency of the proposed evaluation method. Finally, the overall procedure is presented to identify critical variables that can affect the variability of load margins in voltage stability analyses. The proposed method is applied to power systems with a large penetration of renewable power and the influences of critical variabilities on voltage stability are analyzed. In this paper, the proposed method is tested using the IEEE 9-bus and the IEEE 118-bus systems, and its accuracy is compared with those of the local sensitivity analysis method and the commonly used GSA methods.**

***Index Terms*—Renewable power, variability, voltage stability, global sensitivity analysis, stochastic response surface method.**

NOMENCLATURE

Parameters

Y_{ij}	Element of a bus admittance matrix
δ_{ij}	Phase angle of Y_{ij}
G_{ij}	Real part of Y_{ij}
B_{ij}	Imaginary part of Y_{ij}
t_{ij}	Transformation ratio of line ij
μ	Mean
σ	Standard deviation
k, c	Parameters of a Weibull distribution
$r_{\max}$	Maximum solar irradiance
a, b	Parameters of a Beta distribution
C	Copula function of correlated variables
p_i	Probability density function (PDF) of x_i
F_i	Cumulative distribution function (CDF) of x_i
ϕ, Φ	PDF and CDF of a normal variable
ϕ_n, Φ_n	PDF and CDF of a n-dimensional normal variable vector
Σ	Covariance matrix
n	Total number of random input variables
s	Number of variables in the subset y
D	Total variance of f
f_0	Mean of f
N	Sample size of Monte Carlo simulation
a_i	Coefficient of polynomial chaos expansion
S_y	First order global sensitivity index of subset y
$S_{\mathrm{T}z}$	Total global sensitivity index of subset z

Variables

λ	Load margin
V_i	Voltage magnitude of bus i
θ_i	Voltage angle of bus i
$P_{\mathrm{G}i}$, $Q_{\mathrm{G}i}$	Active and reactive power outputs of a conventional generator
$P_{\mathrm{R}i}$, $Q_{\mathrm{R}i}$	Active and reactive power outputs of a renewable generator
$P_{\mathrm{D}i}$, $Q_{\mathrm{D}i}$	Active and reactive power loads at bus i
$b_{\mathrm{P}i}$, $b_{\mathrm{Q}i}$	Directions of load increase at bus i
$P_{\mathrm{L}ij}$, $Q_{\mathrm{L}ij}$	Active and reactive power flow from bus i to bus j
$S_{\mathrm{L}ij,\max}$	Maximum magnitude of apparent power of line ij
x	Input random variable vector
x_i	ith input variable
u_i	Uniform variable in [0, 1]
$\xi_i, \hat{\xi}_i$	Normal variable and standard normal variable
y, z	Subset of x and its complementary subset
γ, α, β	Correlated normal variable vectors
$\hat{\gamma}, \hat{\alpha}, \hat{\beta}$	Independent normal variable vectors
$\bar{z}, \bar{\beta}$	Variable vectors of conditional distributions

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I. Introduction

THE variability of renewable energy has fundamentally changed the planning and operation of electric power systems. Over the last few decades, stochastic power system problems including the probabilistic load flow and probabilistic small signal stability have been studied extensively [1]–[6]. Given the nature of variables, the stochastic methods have aimed at evaluating the uncertain performance of power systems as traditional deterministic methods cannot fully depict the characteristics of variable power systems.

The solution of stochastic power system problems with a large number of random variables will be computationally intractable as we consider all practical variabilities in power systems [7]. However, it may not be necessary to model all such variabilities, in particular those with a minute impact on power system operations. Therefore, accurate identifications of most influencing variables will improve the operational efficiency of power systems with less monitoring, and give a better understanding of relations between input and output variables in solving the stochastic power system problems.

Sensitivity analysis (SA) is a useful tool to evaluate the changes in power system outputs as system inputs are varied. SA provides the information on which input will have the largest effect on designated system outputs. Broadly speaking, SA methods can be divided into two categories: local sensitivity analysis (LSA) and global sensitivity analysis (GSA). LSA usually considers the partial derivative of the output with respect to the input variable. In [8]–[10], the sensitivities of load margins with respective to arbitrary parameters were derived. In [11], the first order sensitivity of the available transfer capacity with respect to power system network variables was described. In [12], a method based on sensitivity analyses was proposed to rank synchronous generators according to their contributions to angular and voltage stability. Although LSA has been widely studied in power system analyses, it also has some shortcomings. Since power systems models are traditionally nonlinear, LSA would only be valid for a small region of variabilities in the vicinity of the power system operating point. Moreover, LSA cannot consider the correlation between different random variables.

Different from LSA, GSA assesses the impact of a full range of possible variabilities and takes into account system nonlinearities [13]–[14]. The variance-based sensitivity analysis, also known as the Sobol method, is one of the most popular GSA methods. By decomposing the variance of the system output into fractions that can be attributed to inputs or sets of inputs, it evaluates the impact of input variables. GSA has been applied to a host of system applications including artificial neural networks [15]–[16] and urban water supplies [17]. However, there have been limited applications of GSA in power systems. Recently, [7] introduced GSA to power distribution system analyses. In [18], several SA methods including LSA, screening SA and GSA were compared for identifying critical parameters that affect the small disturbance stability of a power system.

In previous GSA applications, random input variables were assumed to be independent and statistical correlations of such variables were not considered in evaluating the importance of variabilities. Moreover, several GSA applications to power system analyses suffered from significant computation burdens since the Monte Carlo simulation (MCS) was used in global sensitivity (GS) calculations. Besides, GSA applications to the static voltage stability assessment of power systems with renewable generation were not highlighted.

In this paper, power systems with a significant penetration of renewable generation are studied and an efficient GSA method is proposed to analyze the influence of various variabilities on voltage stability. First, the probabilistic load margin is considered for power systems with renewable generation, and the Copula function is adopted to model the correlations among non-normally distributed variables. Then, GSA for models with correlated variables is presented, and the importance index is defined for input variables. Moreover, the stochastic response surface method (SRSM) is used to establish a surrogate model of the load margin in order to improve the efficiency of calculating sensitivities. Finally, a procedure is designed to identify the important variabilities that affect voltage stability. The proposed method is tested using the IEEE 9-bus and IEEE 118-bus systems, and the results are compared with those of LSA and other GSA methods to validate the effectiveness of the proposed method.

II. Probabilistic Voltage Stability Analysis

The load margin is an effective index for the assessment of power system static voltage stability. The optimization-based method is a popular tool to determine the voltage stability margin [19]–[22]. In this paper, we use an optimal power flow (OPF) model to calculate load margins. The problem aims at maximizing the load margin while satisfying power flow and other operating constraints. The solution to the OPF problem corresponds to the saddle node bifurcation (SNB) point or is determined by operating constraints of power systems [19]. The optimization model is given as follows:

$$\max \qquad \lambda \tag{1.1}$$

subject to:

$$V_i \sum V_j \left|Y_{ij}\right| \cos\left(\theta_{ij} - \delta_{ij}\right) - \left(P_{\mathrm{G}i} + P_{\mathrm{R}i} - P_{\mathrm{D}i} - \lambda b_{\mathrm{P}i}\right) = 0 \tag{1.2}$$

$$V_i \sum V_j \left|Y_{ij}\right| \sin\left(\theta_{ij} - \delta_{ij}\right) - \left(Q_{\mathrm{G}i} + Q_{\mathrm{R}i} - Q_{\mathrm{D}i} - \lambda b_{\mathrm{Q}i}\right) = 0 \tag{1.3}$$

$$P_{\mathrm{G}i,\min} \le P_{\mathrm{G}i} \le P_{\mathrm{G}i,\max} \tag{1.4}$$

$$Q_{\mathrm{G}i,\min} \le Q_{\mathrm{G}i} \le Q_{\mathrm{G}i,\max} \tag{1.5}$$

$$V_{i,\min} \le V_i \le V_{i,\max} \tag{1.6}$$

$$P_{\mathrm{L}ij} = -t_{ij} G_{ij} V_i^2 + V_i V_j \left(G_{ij} \cos\theta_{ij} + B_{ij} \sin\theta_{ij}\right) \tag{1.7}$$

$$Q_{\mathrm{L}ij} = t_{ij} B_{ij} V_i^2 - B_{ij} V_i^2 / 2 + V_i V_j \left(G_{ij} \sin\theta_{ij} - B_{ij} \cos\theta_{ij}\right) \tag{1.8}$$

$$P_{\mathrm{L}ij}^2 + Q_{\mathrm{L}ij}^2 \le S_{\mathrm{L}ij,\max}^2 \tag{1.9}$$

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In the above formulation, the objective is to maximize the power system load margin. Constraints (1.2) and (1.3) are the power balance equations. Constraints (1.4) and (1.5) are the active and reactive power output limits of generators. Constraint (1.6) is the limit of voltage magnitudes. Constraints (1.7)–(1.9) represent the limit of power flow through lines.

In (1), the variability of the direction of active load increase is described by a normal distribution, with the corresponding probability distribution function (PDF) given as

$$p\left(b_{\mathrm{P}i}\right)=\frac{1}{\sqrt{2\pi\sigma_{\mathrm{bP}i}^{2}}}\exp\left(-\frac{\left(b_{\mathrm{P}i}-\mu_{\mathrm{bP}i}\right)^{2}}{2\sigma_{\mathrm{bP}i}^{2}}\right) \tag{2}$$

The reactive power of the load can be described similarly, or designed to keep the power factor constant.

For wind power generation, a Weibull model is commonly used to describe the wind speed with a PDF stated as

$$p\left(v\right)=\left(\frac{k}{c}\right)\left(\frac{v}{c}\right)^{k-1}\exp\left(-\left(\frac{v}{c}\right)^{k}\right),v\geq 0 \tag{3}$$

The photovoltaic (PV) power generation is determined by solar irradiance. The solar irradiance r is described by a Beta distribution:

$$p\left(r\right)=\frac{\Gamma\left(a+b\right)}{\Gamma\left(a\right)\Gamma\left(b\right)}\left(\frac{r}{r_{\max}}\right)^{a-1}\left(1-\frac{r}{r_{\max}}\right)^{b-1} \tag{4}$$

Besides, there exist correlations among uncertain power system variables; for example, adjacent renewable generators will usually have correlated power outputs due to similar weather conditions. Copula is a popular method to establish the joint distribution of multiple variables and Gaussian Copula is selected in this paper to model correlations among non-normally distributed variables [23]–[24]. Note that other Copula functions, such as *t* Copula and Archimedean Copula, can also be used. The Gaussian Copula is stated as

$$C\left(u_1,...,u_n\right)=\Phi_n\left(\Phi^{-1}\left(u_1\right),...,\Phi^{-1}\left(u_n\right);\Sigma\right) \tag{5}$$

where $u_1,\ldots,u_n$ are uniformly distributed random variables in [0, 1], Φ is the cumulative distribution function (CDF) of a normally distributed variable, and Φ_n is an n-dimensional multivariate normal distribution with the covariance matrix Σ.

Here, (1) represents a nonlinear optimization problem with random variables. Given the information on input variables, the probabilistic voltage stability analysis will obtain probability distributions and statistical indices of load margins. Considering a high penetration level of renewable power, we apply the proposed GSA method to identify critical variables among variabilities pertaining to the power system static voltage stability analysis.

III. Global Sensitivity Analysis

A. GSA With Correlated Variables

The probabilistic load margin is described as:

$$Y=f\left(x_1,...,x_n\right) \tag{6}$$

where $x_1,\ldots,x_n$ are n input variables. Consider a subset $y=(x_{i1},\ldots,x_{is})$, $1\leq s<n$, and a complementary subset $z=(x_{j1},\ldots,x_{jn-s})$, such that $x=(y,z)$.

The variance of $f(x_1,\ldots,x_n)$ is decomposed as

$$D=D_y\left[E_z\left[f\left(y,\bar{z}\right)\right]\right]+E_y\left[D_z\left[f\left(y,\bar{z}\right)\right]\right] \tag{7}$$

where $D[\cdot]$ and $E[\cdot]$ represent the variance and the expectation of the function, z and $\bar{z}$ are used to distinguish between a random vector z generated from a joint PDF $p(y,z)$, and a random vector $\bar{z}$ generated from a conditional distribution $p(y,\bar{z}|y)$.

The ratio

$$S_y=\frac{D_y\left[E_z\left[f\left(y,\bar{z}\right)\right]\right]}{D} \tag{8}$$

is known as the first order GS index of the subset y.

Also, the ratio

$$S_{\mathrm{T}z}=\frac{E_y\left[D_z\left[f\left(y,\bar{z}\right)\right]\right]}{D} \tag{9}$$

is known as the total GS index of the subset z.

The index S_y is the faction of the variance of the output that is attributed to the variables in the subset y. Therefore, S_y is used to evaluate a subset of input variables.

The full expression of (8) is

$$S_y=\frac{1}{D}\left[\int_{R^s}\left[\int_{R^{n-s}}f\left(y,\bar{z}\right)p\left(y,\bar{z}\,|y\right)d\bar{z}\right]^2p\left(y\right)dy-f_0^2\right] \tag{10}$$

In the MCS framework the direction application of (8) requires the costly double-loop computation. Hence in [25] a more efficient calculation formula is given as

$$S_y=\frac{1}{D}\left[\int_{R^s}p\left(y\right)dy\left[\int_{R^{n-s}}f\left(y,\bar{z}\right)p\left(y,\bar{z}\,|y\right)d\bar{z}\times\int_{R^{n-s}}f\left(y,\bar{z}'\right)p\left(y,\bar{z}'\,|y\right)d\bar{z}'\right]-f_0^2\right] \tag{11}$$

B. GS Calculation

In this section, we will first deduce the expression of S_y for correlated normal variables and then give the expression of S_y for correlated non-normal variables.

Assume that γ is a vector of n normal variables $\xi_1,\ldots,\xi_n$. $\alpha=(\xi_{i1},\ldots,\xi_{is})$ and is a subset of γ. β is the complementary set of α. Let the mean vectors and covariance matrices of γ, α, β be μ_γ, μ_α, μ_β and Σ_γ, Σ_α, Σ_β, respectively. The mean vector and covariance matrix of γ are partitioned as

$$\mu_\gamma=\begin{bmatrix}\mu_\alpha\\ \mu_\beta\end{bmatrix},\Sigma_\gamma=\begin{bmatrix}\Sigma_\alpha & \Sigma_{\alpha\beta}\\ \Sigma_{\beta\alpha} & \Sigma_\beta\end{bmatrix} \tag{12}$$

For the multivariate normal distribution, the conditional distribution $p(\alpha,\bar{\beta}|\alpha)$ is also a normal distribution:

$$p\left(\alpha,\bar{\beta}\,|\alpha\right)=\phi_{n-s}\left(\alpha,\bar{\beta}\,|\alpha\right) \tag{13}$$

with the mean vector $\mu_{\beta c}$ and the covariance matrix $\Sigma_{\beta c}$ as

$$\begin{aligned}\mu_{\beta c}&=\mu_\beta+\Sigma_{\beta\alpha}\Sigma_\alpha^{-1}\left(\alpha-\mu_\alpha\right)\\ \Sigma_{\beta c}&=\Sigma_\beta-\Sigma_{\beta\alpha}\Sigma_\alpha^{-1}\Sigma_{\alpha\beta}\end{aligned} \tag{14}$$

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Therefore, for correlated normal variables, (11) becomes

$$S_\alpha = \frac{1}{D}\left[\int_{R^s}\phi_s(\alpha)\,d\alpha\left[\int_{R^{n-s}} f(\alpha,\bar{\beta})\,\phi_{n-s}(\alpha,\bar{\beta}\,|\alpha)\,d\bar{\beta} \times \int_{R^{n-s}} f(\alpha,\bar{\beta}')\,\phi_{n-s}(\alpha,\bar{\beta}'\,|\alpha)\,d\bar{\beta}'\right] - f_0^2\right] \quad (15)$$

Furthermore, the PDF of the Gaussian Copula (5) is

$$p(u_1,...,u_n) = \frac{\phi_n(\xi_1,...,\xi_n,\Sigma)}{\prod_{i=1}^n \phi(\xi_i)} \quad (16)$$

where the relation between ξ_i and u_i can be obtained with the cumulative distribution function transformation (CDFT):

$$\xi_i = \Phi^{-1}(u_i) = \Phi^{-1}(F_i(x_i)), i = 1,...,n \quad (17)$$

Applying the transformation law of probabilities (16) and (17), S_y for correlated non-normal variables is [25]

$$S_y = \frac{1}{D}\left[\int_{R^s}\phi_s(\alpha)\,d\alpha\left[\int_{R^{n-s}} f(y,\bar{z})\,\phi_{n-s}(\alpha,\bar{\beta}\,|\alpha)\,d\bar{\beta} \times \int_{R^{n-s}} f(y,\bar{z}')\,\phi_{n-s}(\alpha,\bar{\beta}'\,|\alpha)\,d\bar{\beta}'\right] - f_0^2\right] \quad (18)$$

where

$$y = \left(F_{i_1}^{-1}(\Phi(\xi_{i_1})),...,F_{i_s}^{-1}(\Phi(\xi_{i_s}))\right)$$
$$\bar{z} = \left(F_{j_1}^{-1}\left(\Phi(\bar{\xi}_{j_1})\right),...,F_{j_{n-s}}^{-1}\left(\Phi(\bar{\xi}_{j_{n-s}})\right)\right) \quad (19)$$

The MCS estimation of (18) is

$$S_y = \frac{1}{D}\left[\frac{1}{N}\sum_{k=1}^N\left(f(y_{(k)},z_{(k)})\,f\left(y_{(k)},\bar{z}'_{(k)}\right)\right) - f_0^2\right]$$
$$f_0 = \frac{1}{N}\sum_{k=1}^N f(y_{(k)},z_{(k)})$$
$$D = \frac{1}{N}\sum_{k=1}^N\left(f(y_{(k)},z_{(k)})\right)^2 - f_0^2 \quad (20)$$

where N is the MCS sample size, and subscript "$_{(k)}$" indicates the kth sample of the variable vector.

According to (20), GSA consists of obtaining samples of (y,z), $(y,\bar{z}')$ and evaluating $f(\cdot)$. However, MCS needs a very large sample size to obtain accurate results. In this paper, the stochastic response surface method (SRSM) is adopted for improving the computation efficiency of MCS as we evaluate the power system output for a given set of samples. In the following subsections, we discuss the procedure to generate correlated samples and present SRSM, respectively.

C. Generating Correlated Samples

For correlated variables, the procedure to obtain samples of (y,z) and $(y,\bar{z}')$ is discussed in Procedure 1 and the flowchart is shown in Fig. 1.

Procedure 1: Generating Correlated Samples.

Generating samples of (y,z)
1. Generate two n-dimensional vectors u and u' uniformly distributed in [0, 1]
2. Transform u into an independent standard normal vector w
3. Transform w into a correlated normal vector γ, and split γ into two subsets α and β
4. Obtain (y,z) with CDFT

Generating samples of $(y,\bar{z}')$
5. Transform u' into an independent standard normal vector w′, and split w′ into two subsets $\hat{\alpha}'$ and $\hat{\beta}'$
6. Use (14) to compute $\mu_{\beta c}$ and $\Sigma_{\beta c}$ based on α
7. Perform the Cholesky decomposition $\Sigma_{\beta c} = L_{\beta c}L_{\beta c}^T$
8. Compute $\bar{\beta}' = L_{\beta c}\hat{\beta}' + \mu_{\beta c}$
9. Combine vectors α and $\bar{\beta}'$ to obtain $(\alpha,\bar{\beta}')$
10. Obtain $(y,\bar{z}')$ with (19)

D. SRSM

SRSM is an effective method which is applied to stochastic problems. SRSM will estimate the power system output using the polynomial chaos expansion (PCE) with standard normal variables [21], [26]–[27]. The SRSM steps are given as follows:

1) Transforming random input variables: The power system variabilities, such as wind power, usually follow non-normal distributions. While inputs of SRSM must be standard random variables. Therefore, we first establish the relation between non-normal variables and standard normal variables:

$$x_i = F_i^{-1}\left(\Phi\left(\hat{\xi}_i\right)\right), i = 1,...,n \quad (21)$$

For correlated variables, we apply the method given in Section III-C to obtain standard normal variables.

2) Representing the output with PCE: In this paper, the second-order PCE is used and the output is expressed by:

$$Y = a_0 + \sum_{i=1}^n a_i\hat{\xi}_i + \sum_{i=1}^n a_{ii}\left(\hat{\xi}_i^2 - 1\right) + \sum_{i=1}^n\sum_{j>i}^n a_{ij}\hat{\xi}_i\hat{\xi}_j \quad (22)$$

where a_0, a_i, a_{ii} and a_{ij} are the coefficients of PCE.

3) Estimating PCE coefficients: First, a set of standard normal points is selected. Note that the employed set should contain a sufficient number of points, like 2–3 times the number of coefficients. Second, the samples are transformed into the samples of original variables. Third, deterministic problems are solved to obtain the power system output samples. Finally, the PCE coefficients are estimated by solving a least square problem stated as:

$$\left(H^T H\right)A = H^T I \quad (23)$$

where A is a vector of the coefficients, I is a vector of system output samples, and H is a matrix with the following format:

$$H = \begin{bmatrix} 1 & \hat{\xi}_{1,1} & \cdots & \hat{\xi}_{1,1}^2 - 1 & \cdots & \hat{\xi}_{1,1}\hat{\xi}_{2,1} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \hat{\xi}_{1,K} & \cdots & \hat{\xi}_{1,K}^2 - 1 & \cdots & \hat{\xi}_{1,K}\hat{\xi}_{2,K} & \cdots \end{bmatrix} \quad (24)$$

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Procedure 2: GSA-SRSM.

1. Obtain probability models of random input variables
2. Select a subset of input variables
3. Establish surrogate models of load margins with SRSM
   a. Select points of standard normal variables
   b. Transform the points to correlated samples
   c. Perform deterministic voltage stability analyses
   d. Calculate PCE coefficients
4. Evaluate GSs with MCS
   a. Generate samples of standard normal variables
   b. Evaluate load margins with the surrogate models
   c. Calculate GSs

where K is the number of points, and $\hat{\xi}_{i,j}$ is the jth sample of $\hat{\xi}_i (j = 1, \ldots, K)$.

IV. Procedure of Voltage Stability GSA

The procedure of the GSA application to voltage stability is given in Procedure 2. The key steps are using SRSM to establish surrogate models of load margins and calculating GSs of input variables.

The proposed GSA method considers correlations among input variables. Moreover, the computation efficiency of GSA is improved tremendously by adopting surrogate models of load margins for GS calculation.

V. Case Studies

In this section, the proposed GSA method is tested using the modified IEEE 9-bus and IEEE 118-bus systems. In the IEEE 9-bus system, load variabilities are considered, and the accuracy of the proposed method is compared with those of commonly used methods to validate the proposed approach. In the IEEE 118-bus system, a large penetration of renewables is considered. Critical variabilities affecting the power system voltage stability are identified, and the impacts of such correlations in renewable power generation are analyzed on voltage stability. Programs are developed with Matlab 8.1 on a PC with Inter Core i5 2.40-GHz CPU and 6GB of RAM. Deterministic load margin calculation is solved by the interior point nonlinear optimization method [19].

A. IEEE 9-bus System

In the IEEE 9-bus system, directions of active and reactive load increase are assumed to follow normal distributions. The mean values of normal variables are initial loads, and standard deviations are defined as 10% of the corresponding means. In this IEEE system, the number of input variables is 6 and 56 points are used to establish the SRSM model. The corresponding results are compared with the MCS results using a sample size of 5000.

In Figs. 2 and 3, the probability densities and cumulative distributions of load margins obtained by SRSM are nearly the same as those obtained by MCS. If the same sample size is used in both MCS and SRSM, MCS gives much larger errors.

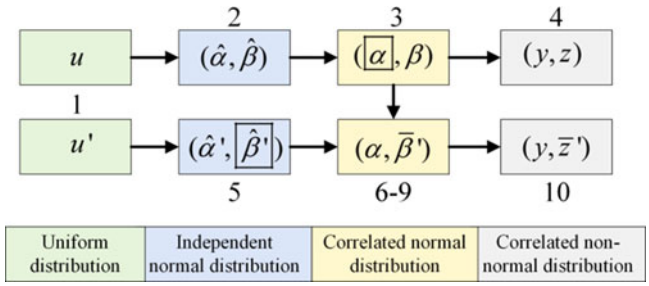


Fig. 1 Flowchart for generating samples of correlated variables.

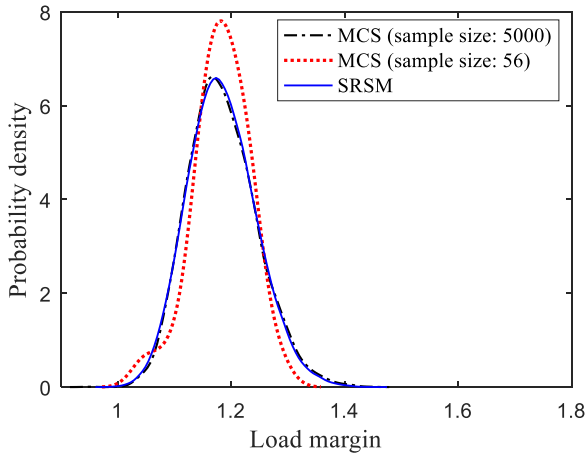


Fig. 2. Probability densities of load margins obtained by SRSM and MCS.

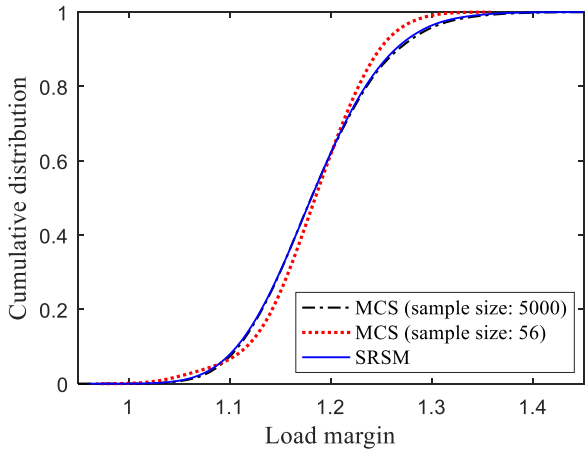


Fig. 3. Cumulative distributions of load margins obtained by SRSM and MCS.

Therefore, SRSM obtains accurate results using much smaller sample sizes.

In the following, GSA is applied to voltage stability using the model established by SRSM. Assume that input variables are independent and the GSA results using the proposed GSA method are compared with those of LSA and the Sobol method. In this case, the LSA index, S_{loc}, determines the impact of each input variable by calculating the partial derivative of the output with respect to the input [7], [9]. The Sobol index, S_{sobol}, uses the method in [14]. In this paper, Sobol sequences [28] with a sample size of 5000 are used to calculate S_{sobol} values. For

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TABLE I
SENSITIVITY INDICES OBTAINED BY LSA AND GSA

Variable	LSA		GSA		
	S_{loc}	Ranking	S_{sobol}	S_y	Ranking
b_{P5}	0.027	2	0.224	0.228	2
b_{P7}	0.022	3	0.180	0.187	3
b_{P9}	0.030	1	0.542	0.543	1
b_{Q5}	0.013	5	0.003	0.005	5
b_{Q7}	0.009	6	0.002	0.003	6
b_{Q9}	0.017	4	0.022	0.026	4

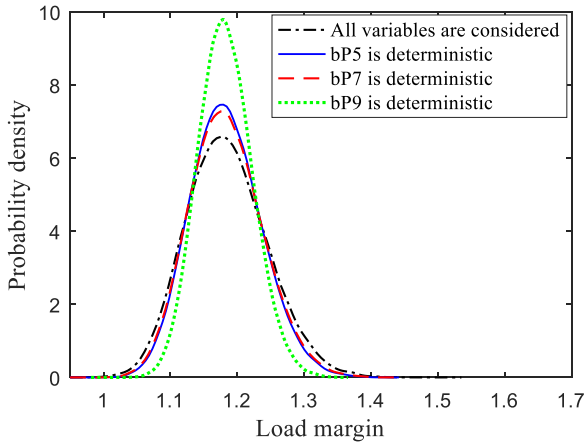


Fig. 4. Probability densities of load margins when load increase directions at some buses are deterministic.

TABLE II
SAMPLE SIZES AND COMPUTATION TIME OF THE PROPOSED METHOD AND SOBOL METHOD

		Sobol method	Proposed method
Establishing	Sample size	–	56
SRSM models	Time (s)	–	4.8
Calculating	Sample size	5000	50000
sensitivities	Time (s)	863.6	0.06
Total computation time (s) [a]		863.6	4.86

[a] Total computation time is the cost of calculating the global sensitivity of one variable.

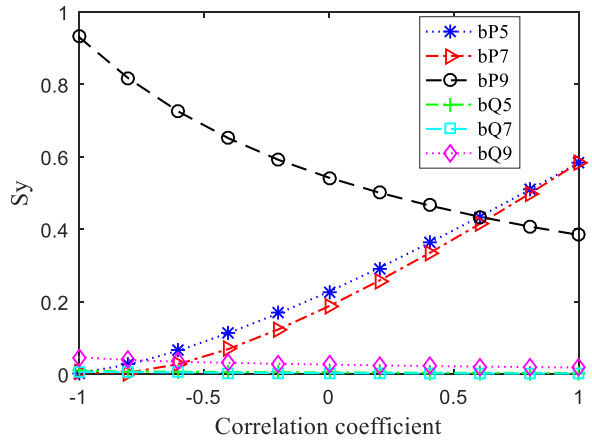


Fig. 5. S_y with respect to correlations.

independent variables, the proposed GSA method and the Sobol method present the same results.

In Table I, the sensitivities of input variables are given. For each variable, S_y is almost equal to S_{sobol}, which demonstrates the accuracy of the proposed GSA results. S_y is different from S_{loc} since they belong to different categories of sensitivity analysis methods. The rankings of variables are the same for LSA and GSA. However, there is a difference between LSA and GSA with respect to the relative importance of variables. For example, according to GSA, S_y of b_{P9} is 0.543, which is much larger than S_y of b_{P5} and b_{P7}, which means that b_{P9} is significantly more important than b_{P5} and b_{P7}. However, according to LSA, the values of S_{loc} for b_{P5}, b_{P7} and b_{P9} are similar. Hence, the three variables have similar influences on the load margin.

In order to analyze the impacts of variables, let b_{P5}, b_{P7} and b_{P9} be deterministic as the probability densities of load margins are compared in Fig. 4. When b_{P5} and b_{P7} are respectively treated as deterministic inputs, the probability distributions of load margins are similar. When b_{P9} becomes deterministic, the probability distribution will be quite different from the other two. This result indicates that b_{P5} and b_{P7} have similar impacts, and b_{P9} has a higher influence than b_{P5} and b_{P7}, which is consistent with the GSA conclusion. Therefore, for independent input variables, compared with LSA, GSA is more accurate to describe the impacts of variables.

Table II shows sample sizes and computation time in the proposed GSA and the Sobol methods. Although the proposed GSA method takes more time than that for establishing SRSM models, the former is still much more efficient since GSs are calculated with the established surrogate model. However, the total computation time of the Sobol method is much longer because it solves the original optimization problem.

As further correlations among variables are considered, the correlation matrix of the Gaussian Copula is assumed to be given by (25) where ρ is in $[-1, 1]$. According to (25), b_{P5} is only correlated with b_{P7} and is independent with other variables. In this problem, LSA and the Sobol method do not work, but the proposed method is still able to perform sensitivity analyses.

$$r = \begin{bmatrix} 1 & \rho & 0 & 0 & 0 & 0 \\ \rho & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{25}$$

Let ρ change from -1 to 1, and the variation of S_y is given in Fig. 5. If b_{P5} and b_{P7} are negatively correlated, S_y of b_{P5} and b_{P7} decrease and S_y of b_{P9} increases as we increase the absolute value of the correlation coefficient. When the correlation coefficient is close to -1, S_y of b_{P5} and b_{P7} will be quite small and S_y of b_{P9} will be close to 1, which indicates that b_{P9} is more effective in calculating the probability distribution of the load margin. If b_{P5} and b_{P7} are positively correlated, S_y of b_{P5} and b_{P7} increase and S_y of b_{P9} decreases as we increase the correlation coefficient. In this system, variabilities of reactive power of loads have a minute influence on load margins.

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TABLE III
PROBABILITY MODELS OF RANDOM INPUT VARIABLES

Model	Wind Weibull	PV Beta	Load Normal
Parameter	$c = 7.41$, $k = 2.06$	$a = 2.06$, $b = 2.5$	$\sigma/\mu = 0.1$
Bus	1, 15, 18, 19, 62, 70, 74, 76, 85	32, 36, 55, 91, 92, 104, 105, 110	Active power load larger than 45 MW

TABLE IV
DESCRIPTION OF DIFFERENT SYSTEMS

System	Uncertainty	Number of variables	
		Normal	Non-normal
S1	Load	33	0
S2	Load and independent renewable power	33	17
S3	Load and correlated renewable power	33	17

In summary, compared with commonly used SA methods, the proposed GSA method is efficient and accurate in evaluating the impact of various variables.

B. IEEE 118-Bus System

In the original IEEE 118-bus system, the total load is 4,242 MW and the total generation capacity is 9,966.2 MW. In order to consider the integration of significant amount of renewable generation, 1,700 MW of conventional generators are replaced by renewable generators, accounting for 40% of the total load. The variabilities of renewable generation are considered and probabilistic models are designed as follows.

The IEEE 118-bus system is divided into four areas of A, B, C and D, covering buses 1–30, 31–60, 61–90 and 91–118, respectively. Conventional generators at buses 1, 15, 18, 19, 32, 36, 55, 62, 70, 74, 76, 85, 91, 92, 104, 105 and 110 are replaced by wind farms or PV power stations with the same capacity of 100 MW. Thus, the renewable generation capacities in areas A, B, C and D are 400 MW, 300 MW, 500 MW and 500 MW, respectively. Three types of random input variables (i.e., wind generation, PV generation and loads) are considered in static voltage stability analyses. The probability models of the variables are given in Table III, and the corresponding parameters are obtained from [26]. Moreover, the renewable units are considered correlated in the same area and independent in different areas. The Gaussian Copula is used to describe correlations among renewable power generation units, and the correlation coefficient of the Gaussian Copula is set as 0.5.

In Table IV, we define three systems to analyze the influence of renewable power on voltage stability. In S1, conventional generators, rather than renewable energy units, are considered, and normally distributed variables are used to describe load variabilities. In S2, the integration of renewable power generation is considered and independent non-normal variables are used to describe renewable power variabilities. In S3, correlated non-normal variables are used to model the dependence among renewable energy units.

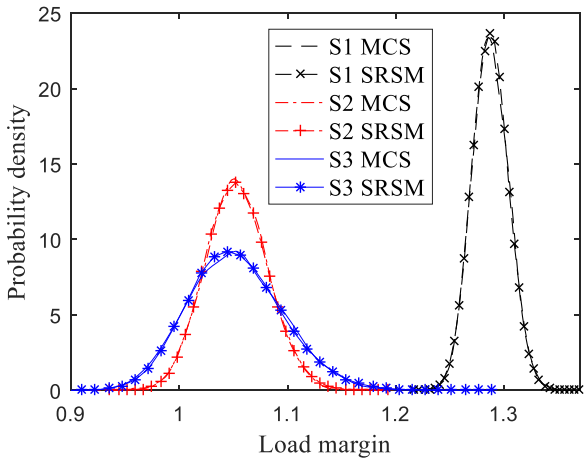


Fig. 6. Probability densities of load margins for different systems.

TABLE V
MEANS AND STANDARD DEVIATIONS OF LOAD MARGINS

	Mean			Standard deviation		
	MCS	SRSM	Error (%)	MCS	SRSM	Error (%)
S1	1.2872	1.2873	0.00	0.0166	0.0167	0.6
S2	1.0527	1.0531	0.04	0.0286	0.0289	1.05
S3	1.0520	1.0523	0.03	0.0430	0.0433	0.70

Both SRSM and MCS are used to obtain the probability distributions and statistical indices of load margins in which the results are compared in Fig. 6 and Table V. In this case, the sample size of MCS is defined as 5000. The probability densities and statistical indices obtained by SRSM are nearly the same as those obtained by MCS. Moreover, the errors of means and standard deviations remain small when correlated non-normal variables are considered in S3, which validates the accuracy of SRSM for large-scale power systems with non-normal distributions.

Although the total generation capacities are the same for all the systems, the distributions of load margins are quite different. For S2 and S3, the means of the load margins are relatively small since the renewable generation is usually smaller than its capacity, and the standard deviations of load margins are relatively large due to the variability of wind and solar power. Besides, correlations among renewable power generation units make the standard deviation larger. Therefore, variabilities make a significant influence on load margins. If traditional deterministic methods are adopted to evaluate the power system voltage stability with renewable power generation, the risk of system collapse cannot be obtained and the voltage stability margin will be overestimated.

There are 33 variables in S1 and 50 variables in S2 and S3. In order to identify important variables affecting load margins, the GSs corresponding to load and renewable power uncertainties are given in Table VI. In S1, there is no renewable generation, and only the load uncertainty determines the variability of load margins. The load uncertainty in area B significantly affects the

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TABLE VI
IMPACTS OF VARIABILITIES IN DIFFERENT AREAS

Variable	System	A	B	C	D
Load	S1	0.091	0.532	0.234	0.129
	S2	0.025	0.098	0.075	0.030
	S3	0.010	0.046	0.035	0.012
Renewable power	S2	0.210	0.087	0.320	0.111
	S3	0.223	0.074	0.414	0.143

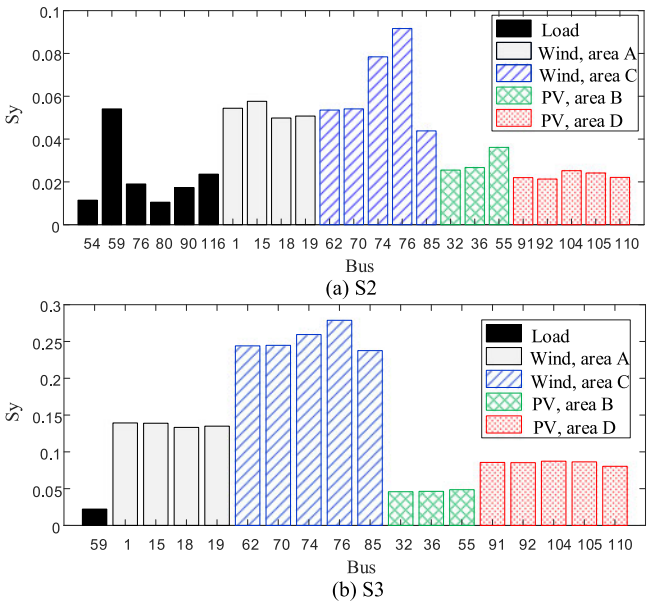


Fig. 7. Variables with S_y larger than 0.01.

load margin, followed by those loads in areas C and D, and the load uncertainty in area A makes a little influence. In S2 and S3, the influences of load variabilities obviously decrease due to much smaller sensitivities, and the variabilities of renewable power become critical factors determining the probability distributions of load margins. In S2 and S3, the GSs of renewable power are 0.730 and 0.863, respectively. Moreover, renewable power participation levels in different areas vary, even though their capacities are similar. The renewable power in area C accounts for most of the variability of the load margin, while the influence of renewable power in area B is much smaller.

In order to fully describe the influence of each input variable, the variables in S2 and S3 with S_y larger than 0.01 are depicted in Fig. 7. In this figure, positive correlations among renewable units make the corresponding S_y larger. Besides, in the same area, sensitivities of renewable power become similar. Moreover, for a subset of correlated variables, the sum of S_y of each variable is larger than S_y of the subset of variables in this system. This is because the attribute of the variable itself and its correlations with other variables are considered in calculating the GS of each input variable. The sum of S_y of each variable will repeatedly calculate the influence of the dependence on system output.

There are 50 variables in S3 and 18 have global sensitivities larger than 0.01. If we merely consider these 18 variables and disregard the rest variables, the probability distribution of the load margin will almost be the same, which is shown in Fig. 8. This result indicates that most of the variables have a minute influence on the load margin, and the proposed method accurately identifies the most effective variables.

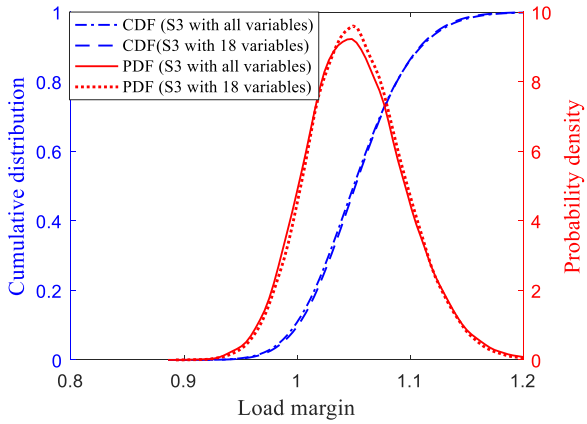


Fig. 8. Probability distributions of load margins for different variables.

TABLE VII
STATISTICAL INDICES OF LOAD MARGINS

	S3 with 18 variables	S3 with all variables	Error (%)
Mean	1.053	1.052	0.10
Standard deviation	0.042	0.043	2.33
5% risk of collapse	0.987	0.985	0.20
25% risk of collapse	1.024	1.022	0.20
75% risk of collapse	1.081	1.080	0.09
95% risk of collapse	1.126	1.128	0.18

The statistical indices of load margins are given in Table VII. Although the standard deviation error is relatively larger than that of the mean, the load margin errors at risks of collapse are quite small, which is consistent with the results given in Fig. 8. These probability distributions and statistical indices of load margins give a comprehensive description of the static voltage stability of power systems with various variabilities. Moreover, the results point out that the influencing variabilities in power system operations which are identified by GSA should be accurately modeled and monitored in order to guarantee the power system stability.

VI. CONCLUSION

In this paper, a GSA method is proposed to consider power system variabilities in the voltage stability evaluation. The proposed method is tested on IEEE 9-bus and IEEE 118-bus systems, and the results are compared with those of other sensitivity analysis methods. The following conclusions are drawn:

1) Compared with LSA, the proposed GSA method is more accurate to consider the relative importance of variables.
2) The results corresponding with the surrogate model of the load margin established by SRSM are accurate compared with those of MCS with a large sample size. With the

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established surrogate model, the computation efficiency of GSA is significantly improved.

3) For power systems with high renewable penetration, the variabilities of renewable power are critical in determining probability distributions of load margins. Moreover, correlations among renewable power affect the variability of load margins.
4) Certain variables affect power system operations significantly. The proposed GSA method accurately identifies such variables and thus can be useful for an effective computation in stochastic power system analyses.

In our future work, we will combine SRSM with the sparse-adaptive scheme to further improve the computation efficiency of GSA in large-scale power system problems. Besides, we will adopt the application of the proposed GSA method in heuristic black-box algorithms such as artificial neural networks to the complicated power system problems in which it is difficult to obtain an analytical closed-form solution.

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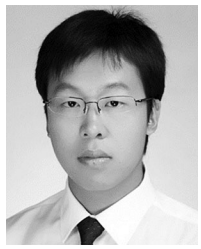
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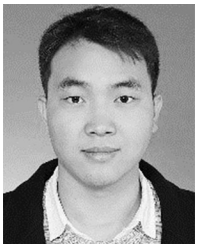
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